

Venkatesh

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PROFESSIONAL SUMMARY

Senior Data Engineer with 14+ years of experience specialized in building end-to-end pipelines and architecting modern data warehouses. Expert in Snowflake and dbt evolution, managing scalable cloud infrastructure (AWS/Terraform), and optimizing processing costs by 35%. Proven ability to translate complex semi-structured data into actionable insights for high-growth AI-driven organizations.

CORE COMPETENCIES

Snowflake Data Warehousing	dbt Modeling & Evolution	Cloud Infrastructure (AWS)	Infrastructure-as-Code (Terraform)
Semi-structured Data Ingestion	Scalable Data Pipelines	Data Governance & Compliance	Apache Airflow / Python

WORK EXPERIENCE

NucleusTeq May 2024 - Present

Senior Data Platform Engineer

- Architecting and delivering scalable **batch and real-time data pipelines** using **Apache Spark** and **Apache Airflow** on **AWS EMR**, processing terabyte-scale datasets.
- Improved distributed data pipeline performance by reducing job execution time from 2-3 hours to 20-30 minutes (**70-85% improvement**) through advanced **Spark tuning** (partitioning, AQE, shuffle optimization).
- Delivered a **zero-downtime, zero-data-loss migration** of a multi-TB scale big data platform for global financial services.
- Architected and supported scalable data platforms processing **10+ TB/day workloads**.
- Implemented **data reliability and observability frameworks**, resulting in a **30-40% reduction in production incidents**.
- Designed optimized analytics-ready data layers using **Spark, Trino, and Iceberg**, improving query performance by **40-60%**.
- Optimized AWS-based infrastructure, achieving **25-35% reduction in cloud compute costs** through fine-grained resource tuning.

Alation May 2023 - Feb 2024

Staff Technical Engineer

- Designed and developed scalable data pipelines for an enterprise-grade **Data Catalog platform**.
- Built **metadata ingestion frameworks** to integrate diverse enterprise data sources.
- Developed advanced **data transformation and enrichment logic** for lineage tracking and metadata standardization.
- Translated business requirements into scalable data models and analytics-ready schemas.

Acceldata Technology Apr 2021 - May 2023

Solutions Engineer

- Contributed to the development of a **data reliability and observability platform**.
- Designed **automated data quality frameworks** (freshness validation, schema drift detection, volume anomaly checks).
- Developed monitoring and alerting mechanisms to detect delayed or missing data.
- Led **root cause analysis** and incident resolution efforts to reduce **MTTD**.

IBM Apr 2017 - Feb 2021

Technology Services Specialist

- Managed and supported distributed data environments, including **cluster configuration** and **performance tuning** in the **Hadoop ecosystem**.
- Developed and maintained **ETL pipelines** for enterprise reporting and analytics within strict SLAs.
- Collaborated with cross-functional teams to troubleshoot production issues and ensure SLA adherence.

Previous Experience

Jun 2011 - Feb 2017

- **Future Focus (Deployed at TCS) | IT Analyst:** Supported enterprise IT infrastructure and resolved technical issues within SLAs.
- **TCS | IT Analyst:** Contributed to data analysis and reporting tasks for international clients.
- **Genpact | Senior Associate:** Handled technical support and data processing for enterprise financial systems.

TOP PROJECTS

Snowflake Data Layer Optimization Hero Metric: 60% Faster Queries

Designed and implemented a high-performance data layer using Snowflake and dbt, enabling real-time analytics for multi-TB datasets.
Tech: Snowflake, dbt, Airflow, SQL

Infrastructure-as-Code Rollout

Standardized cloud-native data platform deployment using Terraform, ensuring consistent and reproducible environments across AWS regions.
Tech: Terraform, AWS, Jenkins, Docker

EDUCATION

B.Sc. in Computer Science 2008
Sri Venkateswara University, India

CERTIFICATIONS

AWS Certified Solutions Architect - Associate 2023
AWS Certified Machine Learning - Specialty 2023

SKILLS

Big Data & Distributed Systems: Hadoop (HDFS, YARN, Hive, HBase, Impala, ZooKeeper), Apache Spark (Spark SQL, AQE, Shuffle Optimization), Apache Iceberg, Trino, Presto

Streaming & Event Processing: Apache Kafka

Cloud Platforms: AWS (EMR, S3, EC2, IAM), Microsoft Azure (HDInsight), Google Cloud Platform (Dataproc, BigQuery)

Data Governance & Security: RBAC, Apache Ranger, Kerberos, Data Quality Frameworks, Monitoring, Observability

Databases & Tools: PostgreSQL, Oracle, MySQL, Airflow, Terraform, Git, CI/CD, Docker, Linux

Programming: Python, SQL, Scala, PySpark, Unix Shell Scripting